# Dependent Type Theory

Charles Averill

The University of Texas at Dallas

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# Where System $F\omega$ Fails

- System $F\omega$ is pretty good, forms the basis of OCaml, Haskell, Rust, C++, Swift type systems
- Point of type systems is "well-typed programs can't 'go wrong' "
- But. . . they still can!

```
let hd (l : 'a list) : 'a =
  match l with
  | [] -> failwith "can't take hd of []"
  | h :: _ -> h
```

# Where System $F\omega$ Fails

What if we could prevent such errors at compile time by extending our type system?

```
let hd (l : 'a (n + 1) list) : 'a =
  match l with
  | h :: _ -> h
```

The type of `hd`'s input argument *depends* on the value `n + 1` to ensure that the list is non-empty.

# Lambda Calculus Refresher

## Syntax

$$e := v \mid \lambda v.\, e \mid e_1\, e_2$$

## Semantics

$$\frac{}{(\lambda v.\, e_1)e_2 \;\Downarrow\; e_1[e_2/v]}(\rightarrow_\beta)$$

$$\frac{P \qquad P \implies Q}{Q}$$

$$(\lambda x.x)(\lambda y.y) \rightarrow_\beta \lambda y.y$$

$$(\lambda x.\lambda y.x)(\lambda a.a)(\lambda b.b) \rightarrow_\beta (\lambda y.(\lambda a.a))(\lambda b.b) \rightarrow_\beta \lambda a.a$$

$$(\lambda f.(\lambda x.f\,x\,x)(\lambda x.f\,x\,x))(\lambda f.(\lambda x.f\,x\,x)(\lambda x.f\,x\,x)) \rightarrow_\beta ?$$

$$\texttt{succ [1,2,3]} \rightarrow_\beta ?$$

# System $F\omega$ Refresher

## Syntax

$$e := () \mid v \mid \lambda v : T.\, e \mid e_1\, e_2$$

$$T := \texttt{unit} \mid \texttt{void} \mid T_1 \to T_2$$

## Typing Semantics

$$\frac{x : \sigma \in \Gamma}{\Gamma \vdash x : \sigma} \qquad \frac{\Gamma, x : \sigma \vdash e : \tau}{\Gamma \vdash (\lambda x : \sigma.\, e) : (\sigma \to \tau)} \qquad \frac{\Gamma \vdash e_1 : \sigma \to \tau \quad \Gamma \vdash e_2 : \sigma}{\Gamma \vdash e_1\, e_2 : \tau}\ (\lambda_\to)$$

$\Gamma$: *typing context*, function that maps idents to types

$\Gamma \vdash e : \tau$: "in context $\Gamma$, $e$ has type $\tau$"

$$\lambda x : \texttt{int}.\lambda y : \texttt{int}.\, x + y$$

$$(\lambda s : \texttt{str}.\, \texttt{"Hello, "} + s + \texttt{"!"})\, \texttt{"World"}$$

# System $F\omega$ Refresher

## Syntax

$$e := () \mid v \mid \lambda v : T.\, e \mid e_1\, e_2 \mid \Lambda\alpha.\, e$$

$$T := \texttt{unit} \mid \texttt{void} \mid T_1 \to T_2 \mid \alpha \mid \forall\alpha.\, T$$

## Typing Semantics

$$\frac{x : \sigma \in \Gamma}{\Gamma \vdash x : \sigma} \qquad \frac{\Gamma, x : \sigma \vdash e : \tau}{\Gamma \vdash (\lambda x : \sigma.\, e) : (\sigma \to \tau)} \qquad \frac{\Gamma \vdash e_1 : \sigma \to \tau \quad \Gamma \vdash e_2 : \sigma}{\Gamma \vdash e_1\, e_2 : \tau} \; (\lambda_\to)$$

$$\frac{\Gamma, \alpha \vdash e : \sigma}{\Gamma \vdash \Lambda\alpha.\, e : \forall\alpha.\, \sigma} \qquad \frac{\Gamma \vdash e : \forall\alpha.\, \sigma}{\Gamma \vdash e\,[T] : \sigma[T/\alpha]} \; (\lambda_2)$$

$$\textsc{Cons} : \forall\alpha.\, \alpha \to \alpha\ \texttt{list} \to \alpha\ \texttt{list} := \Lambda\alpha.\, \lambda hd : \alpha.\lambda tl : \alpha\ \texttt{list}.\, (hd,\, tl)$$

# System $F\omega$ Refresher

## Syntax

$$e := () \mid v \mid \lambda v : T.\ e \mid e_1\ e_2 \mid \Lambda\alpha :: K.\ e \mid e\ [T]$$

$$T := \texttt{unit} \mid \texttt{void} \mid T_1 \to T_2 \mid \alpha \mid \forall\alpha :: K.\ T \mid \lambda\alpha :: K.\ T \mid T_1\ T_2$$

$$K := * \mid K_1 \to K_2$$

## Typing Semantics (assuming type equivalence via $\beta$-reduction)

$$\frac{x : \sigma \in \Gamma}{\Gamma \vdash x : \sigma} \qquad \frac{\Gamma, x : \sigma \vdash e : \tau}{\Gamma \vdash (\lambda x : \sigma.\ e) : (\sigma \to \tau)} \qquad \frac{\Gamma \vdash e_1 : \sigma \to \tau \quad \Gamma \vdash e_2 : \sigma}{\Gamma \vdash e_1\ e_2 : \tau}\ (\lambda_\to)$$

$$\frac{\Gamma, \alpha \vdash e : \sigma}{\Gamma \vdash \Lambda\alpha.\ e : \forall\alpha.\ \sigma} \qquad \frac{\Gamma \vdash e : \forall\alpha.\ \sigma \quad \Gamma \vdash T :: \kappa}{\Gamma \vdash e\ [T] : \sigma[T/\alpha]}\ (\lambda_2) \qquad \frac{\alpha :: \kappa \in \Gamma}{\Gamma \vdash \alpha :: \kappa} \qquad \frac{t \in \lambda_2}{\Gamma \vdash t :: *}$$

$$\frac{\Gamma, x :: \kappa_1 \vdash t :: \kappa_2}{\Gamma \vdash (\lambda x :: \kappa_1.\ t) :: (\kappa_1 \to \kappa_2)} \qquad \frac{\Gamma \vdash t_1 :: \kappa_1 \to \kappa_2 \qquad \Gamma \vdash t_2 :: \kappa_1}{\Gamma \vdash t_1\ t_2 :: \kappa_2}\ (\lambda_{\underline{\omega}})$$

# System $F\omega$ Refresher

$$\textsc{Pair} :: * \to * \to * :=$$
$$\lambda A :: *. \lambda B :: *. \forall F. (A \to B \to F) \to F$$
$$\texttt{tuple} : \forall A. \forall B.\ A \to B \to \textsc{Pair}\ A\ B :=$$
$$\Lambda A. \Lambda B. \lambda fst : A. \lambda snd : B. \Lambda F. \lambda f : (A \to B \to F). f\ fst\ snd$$
$$\textsc{Sum} :: * \to * \to * :=$$
$$\lambda A :: *. \lambda B :: *. \forall F. (A \to F) \to (B \to F) \to F$$
$$\texttt{left} : \forall A. \forall B.\ A \to \textsc{Sum}\ A\ B :=$$
$$\Lambda A. \Lambda B. \lambda a : A. \Lambda F. \lambda f_1 : (A \to F). \lambda f_2 : (B \to F). f_1\ a$$
$$\texttt{right} : \forall A. \forall B.\ B \to \textsc{Sum}\ A\ B :=$$
$$\Lambda A. \Lambda B. \lambda b : B. \Lambda F. \lambda f_1 : (A \to F). \lambda f_2 : (B \to F). f_2\ b$$

# Type Inhabitation

A type is *inhabited* iff there exists a term with that type.

| Type | Inhabited? |
|---|---|
| $\texttt{unit}$ | $()$ |
| $\texttt{void}$ | $\times$ |
| $\texttt{unit} \to \texttt{unit}$ | $\lambda x : \texttt{unit}.\ x$ |
| $\texttt{unit} \to \texttt{void}$ | $\times$ |
| $\texttt{void} \to \texttt{unit}$ | $\lambda x : \texttt{void}.\ ()$ |
| $\texttt{void} \to \texttt{void}$ | $\lambda x : \texttt{void}.\ x$ |
| $\forall \alpha :: *.\ \alpha \to \alpha$ | $\Lambda \alpha :: *.\ \lambda x : \alpha.\ x$ |
| $\forall \alpha :: *.\ \alpha$ | $\times$ |
| $\forall \alpha :: *.\ \forall \beta :: *.\ \alpha \to \beta$ | $\times$ |
| $(\lambda \beta :: *.\ \beta \to \beta)\ \texttt{unit}$ | $\lambda x : \texttt{unit}.\ x$ |
| $\forall F :: * \to *.\ \forall \alpha :: *.\ F\ \alpha \to F\ \alpha$ | $\Lambda F :: * \to *.\ \Lambda \alpha :: *.\ \lambda x : F\ \alpha.\ x$ |

# Type Inhabitation

We can fill out this relation with some added standard types, like tuples:

| Type | Inhabited |
|---|---|
| `unit` | True |
| `void` | False |
| `void` $\times$ `void` | False |
| `void` $\times$ `unit` | False |
| `unit` $\times$ `void` | False |
| `unit` $\times$ `unit` | True |
| `void` $\rightarrow$ `void` | True |
| `void` $\rightarrow$ `unit` | True |
| `unit` $\rightarrow$ `void` | False |
| `unit` $\rightarrow$ `unit` | True |
| $\forall \alpha :: *.\ \alpha$ | False |
| $\forall \alpha :: *.\ \alpha \rightarrow \alpha$ | True |

# Type Inhabitation

We can fill out this relation with some added standard types, like tuples:

| Logical Expression | Truth Value |
|---|---|
| $T$ | True |
| $F$ | False |
| $F \wedge F$ | False |
| $F \wedge T$ | False |
| $T \wedge F$ | False |
| $T \wedge T$ | True |
| $F \to F$ | True |
| $F \to T$ | True |
| $T \to F$ | False |
| $T \to T$ | True |
| $\forall P.\ P$ | False |
| $\forall P.\ P \to P$ | True |

Break! Any Questions?

# Types as Propositions

It turns out that System $F\omega$ corresponds with Higher-Order Propositional Logic (HOL)! $F\omega$ types are propositions, and its values are proofs of propositions.

```
let modus_ponens : P -> (P -> Q) -> Q =
  fun (p : P) (f : P -> Q) ->
    f p
```

$$\frac{P \quad P \implies Q}{Q}(\text{MP})$$

What do we get with higher-order propositional logic?

- Propositions that talk about other propositions (reasoning about logical connectives):

$$\textsc{Symmetric}(F) \coloneqq \forall P\, Q,\ F\, P\, Q \implies F\, Q\, P$$

- Encodings of impredicative relations and data structures:

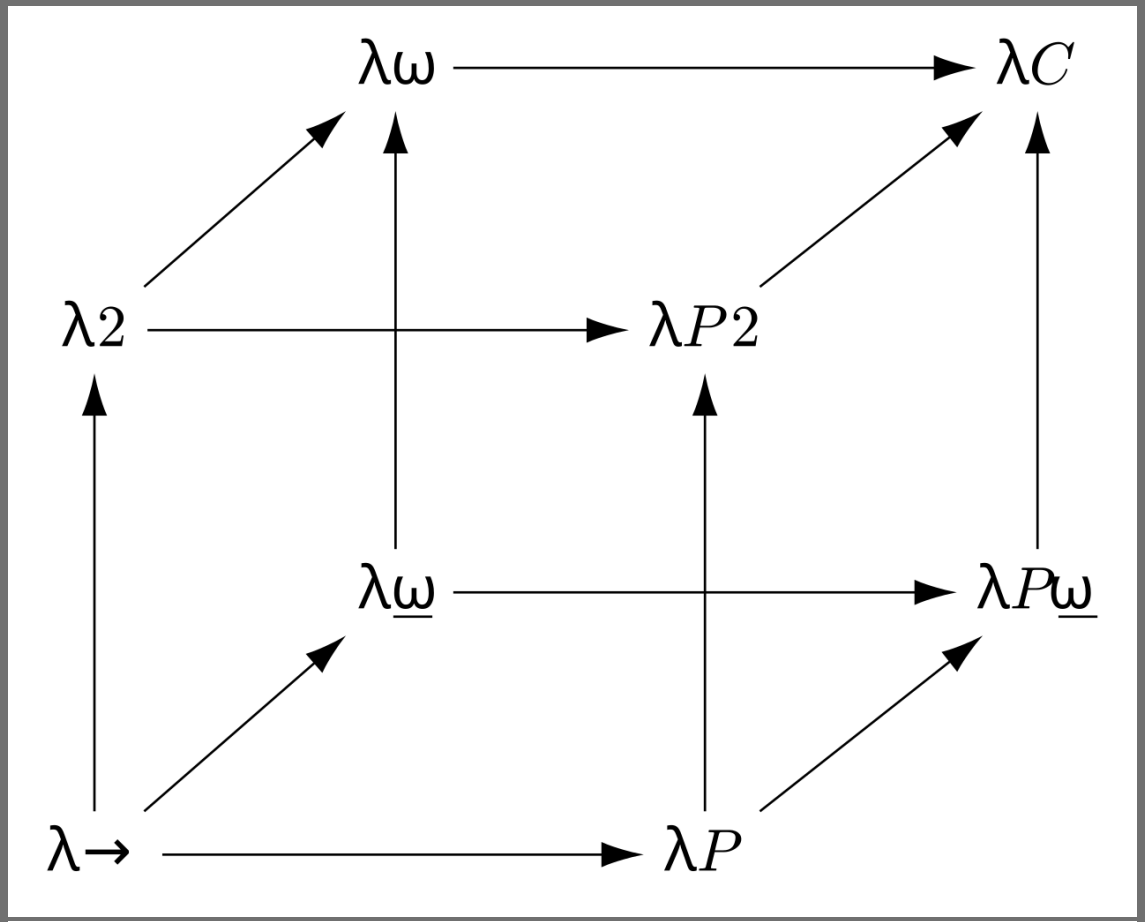
$$\textsc{Eq}(P,\ Q) \coloneqq \forall R,\ R\, P \implies R\, Q$$

# Where it Stops

- HOL is pretty nice (see Isabelle/HOL), but has limitations
- Can't even define predicates over values, like `5 is odd`
- HOL propositions can only take *other propositions* as parameters
- Not even induction is native to HOL!

# Back to PL

The problem is related to the lambda calculi typing semantics!



# $\lambda C$

## Syntax

$$e \coloneqq \lambda_\omega$$
$$T \coloneqq \lambda_\omega \mid \Pi v : T_1 . T_2$$
$$K \coloneqq \lambda_\omega \mid \Pi v : T . K$$

## Typing Semantics (plus $\lambda_\omega$)

$$\frac{\Gamma \vdash T_1 :: * \qquad \Gamma, x : T_1 \vdash T_2 :: *}{\Gamma \vdash \Pi x : T_1 . T_2 :: *} \qquad \frac{\Gamma \vdash S :: \Pi x : T . K \qquad \Gamma \vdash t : T}{\Gamma \vdash S\, t :: K[t/x]}$$

$$\frac{\Gamma \vdash S :: * \qquad \Gamma, x : S \vdash t : T}{\Gamma \vdash (\lambda x : S . t) : \Pi x : S . T} \qquad \frac{\Gamma \vdash t_1 : \Pi x : S . T \qquad \Gamma \vdash t_2 : S}{\Gamma \vdash t_1\, t_2 : T[t_2/x]}$$

$$\frac{\Gamma \vdash T :: * \qquad \Gamma, x : T \vdash S :: K}{\Gamma \vdash (\lambda x : T . S) :: \Pi x : T . K} \qquad \frac{\Gamma \vdash T :: * \qquad \Gamma, x : T \vdash K}{\Gamma \vdash \Pi x : T . K} \; (\lambda C)$$

# Dependent Vectors

$$\textsc{Vec} :: * \to \Pi n : \texttt{nat}. * :=$$
$$\lambda T :: *. \lambda n : \texttt{nat}. n = 0 \; ? \; \texttt{unit} : \textsc{Pair}\; T\; (\textsc{Vec}\; T\; (n-1))$$
$$\texttt{nil} : \forall T :: *. \textsc{Vec}\; T\; 0 := \Lambda T :: *. ()$$
$$\texttt{cons} : \forall T :: *. \Pi n : \texttt{nat}. T \to \textsc{Vec}\; T\; n \to \textsc{Vec}\; T\; (n+1) :=$$
$$\Lambda T :: *. \lambda n : \texttt{nat}. \lambda hd : T. \lambda tl : \textsc{Vec}\; T\; n. \texttt{tuple}\; [T]\; [\textsc{Vec}\; T\; n]\; hd\; tl$$
$$\texttt{hd} : \forall T :: *. \Pi n : nat. \textsc{Vec}\; T\; (n+1) \to T :=$$
$$\texttt{fst}\; [T]\; [\textsc{Vec}\; T\; n]$$
$$\texttt{tl} : \forall T :: *. \Pi n : nat. \textsc{Vec}\; T\; (n+1) \to \textsc{Vec}\; T\; n :=$$
$$\texttt{snd}\; [T]\; [\textsc{Vec}\; T\; n]$$